REVIEW

The onset of sarcoidosis in relation to tattoos: Focus on the treatment landscape. a critical literature review and case report

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ABSTRACT

Sarcoidosis is a complex multisystem granulomatous disease characterized by the formation of non-caseating granulomas in affected tissues, primarily involving the lungs and the lymphatic system. While its precise etiology remains elusive, a combination of genetic predisposition, environmental factors, and immune dysregulation is believed to contribute to disease development. In recent years, an intriguing association has emerged between cutaneous sarcoidosis and tattooing. Several case reports and observational studies have documented instances where sarcoid granulomas develop within tattooed skin, often with concurrent systemic involvement. This review explores the existing literature on the relationship between tattoos and sarcoidosis, examining proposed pathogenic mechanisms, clinical presentations, and implications for diagnosis and management. The evidence suggests that tattoo pigments—particularly those containing metals and organic compounds—may act as foreign bodies, eliciting a localized granulomatous immune response that can potentially trigger or reflect systemic disease. Although causality remains unproven, understanding this association is vital for clinicians managing patients with granulomatous skin lesions and for public health considerations related to tattoo safety. Further research is

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necessary to clarify the pathophysiological links and to identify individuals at risk, ultimately guiding preventive strategies and tailored therapeutic approaches.

Key words: sarcoidosis, tattoo sarcoidosis, cutaneous sarcoidosis, immunosuppressive therapies, skin biopsy

Introduction

Sarcoidosis is a complex multisystem granulomatous disease that can affect virtually any organ, with the lungs, lymph nodes, skin, and eyes being the most frequently involved (1,2). Histologically, the disease is defined by the presence of non-caseating granulomas—organized collections of epithelioid macrophages and multinucleated giant cells—without the central caseation necrosis characteristic of other granulomatous conditions (1).

The overall incidence varies substantially by geographic region and ethnicity, with the highest rates reported in Scandinavian and African-American populations; globally, incidence estimates range from 1 to 35 per 100,000 persons per year (1,2). Despite decades of research, the precise etiology of sarcoidosis remains elusive. The disease is currently believed to result from an exaggerated Th1-polarized immune response to environmental, infectious, or self-antigens in genetically predisposed individuals, leading to granuloma formation as the body attempts to wall off perceived foreign material (1,3).

In recent years, growing attention has been directed toward the potential role of tattoos as environmental triggers or local catalysts for granuloma formation. As tattooing has become increasingly prevalent worldwide, cases have emerged suggesting a possible link between tattoo inks and the development of sarcoid-like reactions (4,5). Tattoo inks—especially those containing inorganic metals such as titanium dioxide, nickel, cobalt, or certain azo dyes—can provoke localized granulomatous responses that may mimic sarcoidosis or coexist with systemic disease, substantially complicating diagnosis and management (6,7). Documented instances exist in which granulomas initially confined to tattooed skin subsequently progressed to involve systemic organs such as the lungs and lymph nodes, raising questions about the potential initiating role of tattoo pigments (8).

The underlying pathogenesis is thought to involve persistent foreign body reactions to pigment particles introduced during tattooing. These particles act as chronic antigens, triggering hypersensitivity responses and immune dysregulation in susceptible individuals (6,7). Immune activation may begin as a localized cutaneous reaction but, in some cases, progress to systemic sarcoidosis through mechanisms involving immune cell migration and cytokine-mediated inflammation, a process likely modulated by genetic factors that predispose certain individuals to exaggerated immune responses to foreign materials (1,3).

Clinically, tattoo-associated sarcoidal reactions typically present as firm nodular lesions within tattooed areas, appearing months or even years after the initial procedure (5,6). Lesions may manifest as isolated skin nodules or coincide with systemic symptoms, and can easily be mistaken for benign cosmetic reactions or uncomplicated allergic responses, delaying appropriate diagnosis and management (4,7). Recognizing these presentations is clinically important, as they may represent the first manifestation of underlying systemic sarcoidosis (8,9).

Diagnosis requires a combination of histopathological analysis and systemic investigation. A skin biopsy revealing non-caseating granulomas is essential for confirmation but must be carefully differentiated from infectious causes such as tuberculosis or fungal infections (2,7). Additional systemic assessments include chest imaging and laboratory tests, including serum angiotensin-converting enzyme (ACE) levels and 24-hour urinary calcium, which may serve as additional biomarkers of disease activity and organ involvement to evaluate the extent of organ involvement (1,2).

Management strategies typically involve immunosuppressive therapies, with systemic corticosteroids representing the established first-line treatment (1,2). In refractory or corticosteroid-dependent cases, steroid-sparing agents such as methotrexate, hydroxychloroquine, and increasingly biologic therapies are employed (10-13). Preventive measures focus on the regulation of tattoo ink composition and on increased awareness among clinicians and the public regarding the potential immunological risks associated with tattooing (4,5).

While a direct causal relationship between tattoos and sarcoidosis has not yet been conclusively established, accumulating clinical and histological evidence suggests that tattoos can serve as sites or triggers for granulomatous reactions, potentially contributing to systemic disease in predisposed individuals (1,6,8). This critical literature review, integrated with a representative case report from our centre, is specifically focused on the treatment landscape of tattoo-associated sarcoidosis (TS).

Methods

This critical literature review, integrated with a case report, was conducted by analyzing original articles, narrative and systematic reviews, clinical guidelines, and case reports published up to April 2026. Narrative reviews were included selectively when they provided expert synthesis of diagnostic or therapeutic approaches not yet covered by primary studies, and are clearly identified as such in the text; they were not used as primary sources of novel data. The primary electronic databases searched were PubMed, Scopus, and Web of Science, using the following keywords: "tattoo sarcoidosis", "tattoo sarcoidosis biopsy", "tattoo sarcoidosis treatment", "tattoo sarcoidosis glucocorticoids", and "tattoo sarcoidosis immunosuppression". Clinical studies, narrative reviews, and case reports reporting data on the diagnosis, treatment, and prognosis of TS were considered for inclusion. Inclusion criteria comprised: 1) studies on patients with a confirmed diagnosis of TS; 2) studies describing the use of skin biopsies for the diagnosis and monitoring of TS; and 3) studies providing details on the therapeutic regimens employed. Studies without specific data on TS, studies on unrelated pathologies, and unpublished abstracts were excluded. All diagnostic and therapeutic procedures in the case report were performed in accordance with current ethical and clinical guidelines [14 relevant articles were selected: 5 case reports, 2 observational studies, 5 reviews, 2 case reports with review] (14-25).

Case report

This case describes a middle-aged Romanian man with a long-standing history of pulmonary sarcoidosis, initially diagnosed in 2014 via transbronchial biopsy. At that time, he was treated with oral prednisone starting at 25 mg daily, which was gradually tapered and discontinued by 2015. Over subsequent years, his pulmonary disease progressed to a fibrotic stage characterized by irreversible scarring and reduced lung compliance. Despite this, he achieved periods of clinical stability on combined corticosteroid and methotrexate therapy, which helped to control active systemic inflammation.

In addition to his pulmonary condition, the patient has a significant dermatological history. At approximately 1.5 years of age, he sustained skin burns requiring split-thickness grafting. Over time, ulcerations developed at the grafted sites. During one episode, a skin biopsy was performed; the wound subsequently became colonized with bacteria, requiring treatment first with topical antibiotics and later with a course of systemic antibiotic therapy. Following resolution of the infection, the lesion healed with scar formation. In 2013, the patient underwent extensive tattooing over his entire back (Figure 1). His medical history also includes previous hepatitis B and C infections, both successfully managed with antiviral therapy, with no current active hepatic disease.

It is worth noting that the cutaneous sarcoid lesions in this patient developed at sites of both preexisting burn scarring and tattoo pigmentation, raising the question of whether the primary risk factor was the underlying scar tissue rather than the tattoo per se. Indeed, scar tissue is a well-recognized site of sarcoid infiltration (so-called scar sarcoidosis), as the local inflammatory microenvironment may represent

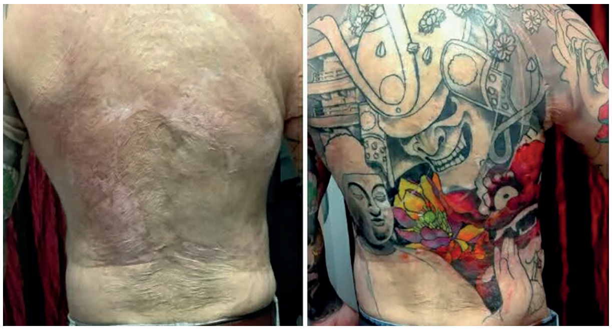

Figure 1. Patient's back before (left) and after (right) extensive tattooing in 2013. The pre-existing burn scar graft site is visible at the centre of the back. This site subsequently developed sarcoid infiltration and ulceration, illustrating the potential contribution of both scar tissue and tattoo pigment as co-localizing triggers for cutaneous granuloma formation.

a predisposed substrate for granuloma formation. In this case, the temporal proximity of tattooing over the scarred area and the subsequent development of cutaneous sarcoidosis suggests that both factors may have contributed synergistically; particularly, the scar tissue provided a pre-existing inflammatory milieu and the tattoo pigment acted as an additional persistent antigenic stimulus. This distinction is clinically relevant and warrants consideration in future reports of tattoo-associated sarcoidosis presenting at sites of prior cutaneous trauma.

At the time of evaluation, the patient reported increased fatigue, dry mouth, diffuse polyarthralgia involving multiple joint districts, occasional dry eye symptoms, and a bronchial exacerbation managed conservatively. His current medication regimen includes low-dose methylprednisolone (4 mg daily), methotrexate (7.5 mg weekly), and inhaled bronchodilators for airway management.

Pulmonary assessments revealed a slight decline in lung function, with decreased diffusing capacity for carbon monoxide (DLCO) and a trend toward restrictive patterns on spirometry compared with prior assessments performed at disease diagnosis in 2014. Despite these functional changes, the patient remained clinically stable without significant respiratory symptoms at rest. PET-CT imaging demonstrated persistent but reduced metabolic activity in the lungs and mediastinal lymph nodes, consistent with partial remission of active granulomatous disease.

Laboratory evaluations revealed lymphopenia, particularly affecting CD4+ T cells, and mild hypercalciuria. Serum sIL-2R was not evaluated in our patient. Biochemical parameters and arterial blood gases remained within normal limits. Ophthalmological examination was unremarkable, with no signs of ocular involvement.

On physical examination, residual ulcerations at prior burn and tattoo sites were identified, prompting a skin biopsy to assess for possible sarcoid infiltration. Histopathological analysis confirmed the presence of non-caseating granulomas consistent with cutaneous sarcoidosis. Given the biopsy findings and evidence of ongoing systemic inflammatory activity, the methylprednisolone dose was increased from 4 mg to 6 mg daily. Topical corticosteroid therapy was initiated for the cutaneous lesions, and a short course of proton pump inhibitor therapy was started to address suspected gastroesophageal reflux. Methotrexate was maintained at the current dose for ongoing systemic disease control. Systemic involvement in our patient extended beyond the skin and lungs, including a notable history of gastrointestinal manifestations, a well-recognized but often underdiagnosed feature of sarcoidosis.[26] Autoantibody profiling, increasingly recognized as a potential biomarker for organ involvement in sarcoidosis, was also considered as part of the diagnostic workup (27).

At one-month telephone follow-up, the patient reported clear improvement of the cutaneous lesions with topical corticosteroid therapy. It should be noted, however, that the methylprednisolone dose was simultaneously increased from 4 to 6 mg daily at the time topical therapy was initiated; consequently, it cannot be excluded that the improvement in cutaneous lesions was attributable in part or in full to the intensification of systemic corticosteroid therapy rather than exclusively to the topical treatment. Systemic symptoms, including fatigue and polyarthralgia, had also improved in parallel with the adjustment of systemic therapy, and repeat laboratory parameters showed favorable trends. No other tattoo sites or scar areas on the patient's body were affected by sarcoid infiltration at this assessment. Regular follow-up was planned, including laboratory

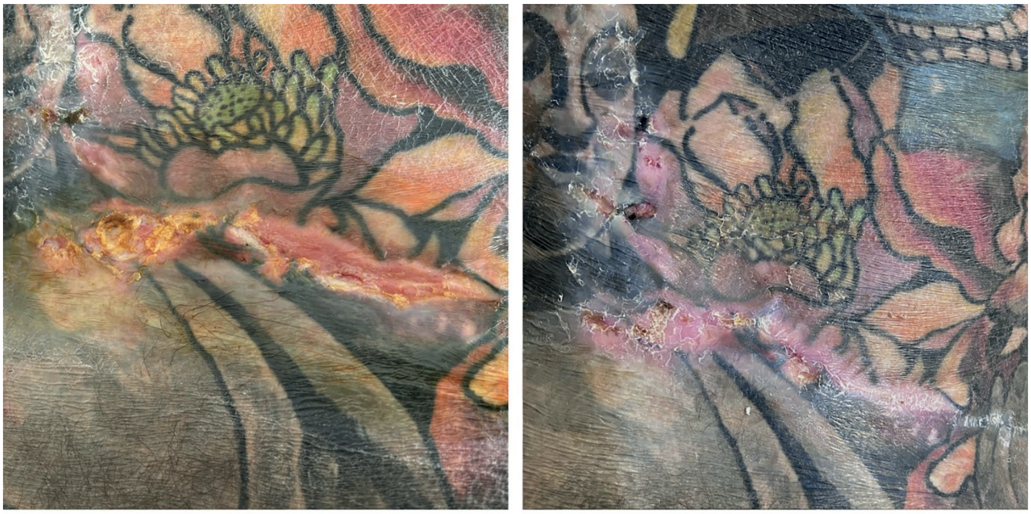

Figure 2. Close-up comparison of the cutaneous sarcoid lesion at the tattoo site. Pre-treatment (left): active ulceration with fibrinous exudate and surrounding granulomatous tissue. Post-treatment (right): significant reduction in ulcer depth and inflammatory activity. Note that concurrent intensification of systemic corticosteroid therapy may have contributed to the observed improvement.

tests, pulmonary function testing, and imaging, to monitor disease course and treatment response.

This case illustrates the importance of a vigilant, multidisciplinary approach in patients with pre-existing systemic sarcoidosis who develop new cutaneous manifestations at tattoo sites. The successful response to topical corticosteroids (or to the concurrent increase in systemic corticosteroid dose) highlights the value of early biopsy-confirmed diagnosis in guiding targeted therapeutic decisions, while the concurrent optimization of systemic therapy underscores the need for integrated management of both cutaneous and systemic disease components (Figure 2). Notably, microvascular assessment via nailfold capillaroscopy and other tools may offer additional insight into systemic disease activity such as sarcoidosis,[28,29] and could be considered in future evaluations of this patient.

Discussion

The prevalence and true extent of tattoo-related skin complications remain incompletely understood. Kluger estimated that complications from tattooing—ranging from allergic reactions to granulomatous disease—are more common than generally recognized, though precise incidence data are lacking due to significant underreporting (4,5). Kazandjieva et al. estimated that approximately 2% of tattooed individuals experience some form of skin reaction, while Klügl et al. identified persistent skin issues in approximately 6% of tattooed persons, including chronic dermatitis and granulomatous reactions. Høgsberg et al. reported that minor complaints such as pruritus, dryness, or mild inflammation affect up to 27% of tattooed individuals in Denmark, with symptoms often persisting beyond three months (4). In the absence of routine histopathological examination, the exact nature of these reactions remains uncertain, making it difficult to distinguish benign irritations from clinically significant pathological processes (7).

Although relatively rare, tattoo-associated reactions may occasionally reveal underlying systemic conditions such as sarcoidosis. The estimated incidence of sarcoidosis presenting with cutaneous manifestations triggered by tattoos is approximately 1–2 cases per 10,000 Caucasian individuals (4,5). Many cases likely go unrecognized, as patients may be reluctant to seek medical attention for mild or non-specific symptoms, incomplete biopsies may yield inconclusive results, and granulomatous inflammation can be misinterpreted as allergic contact dermatitis (7). Differentiating between foreign-body granulomas and true sarcoidosis is particularly challenging when only a single tattoo color is involved, and special staining techniques combined with careful clinicopathological correlation are often required (6,7).

Epidemiologically, tattoo sarcoidosis tends to affect middle-aged men more frequently, with a male-to-female sex ratio of approximately 2.9:1 (9). Although women are generally more likely to have tattoos, the male predominance in TS suggests possible hormonal or immunological modulating factors. With regard to ethnicity, while sarcoidosis in general is known to be more common and to follow a more severe course in individuals of African descent, specific data on the relative incidence or severity of tattoo-associated sarcoidosis by ethnicity are limited, and no published study has formally compared TS rates or outcomes between ethnic groups; this remains an important area for future investigation. Skin lesions typically present as scattered, well-defined papules or nodules within the tattooed areas, and systemic cutaneous involvement outside tattooed regions such as at sites of prior trauma, as observed in our patient is infrequent but documented (6,8).

Given the potential for systemic involvement, long-term monitoring is essential. Approximately 30% of individuals with tattoo-related sarcoidosis may subsequently develop systemic disease, which may initially be subclinical and detectable only through routine investigations such as chest imaging (9). A comprehensive baseline evaluation—including chest X-ray or high-resolution computed tomography and laboratory tests—should therefore be routinely performed in all patients presenting with granulomatous tattoo reactions (1,2). Sarcoidosis with systemic involvement may additionally affect the gastrointestinal tract, a manifestation that is often underrecognized and underdiagnosed in clinical practice (26). Autoantibodies have also been proposed as potential biomarkers for organ involvement in sarcoidosis (27), and emerging evidence suggests that nailfold capillaroscopy may provide a non-invasive window into microvascular disease activity (28).

The immunopathogenesis of TS remains incompletely understood. Environmental factors including inorganic metals and chemical components of tattoo inks are hypothesized to act as persistent antigens in genetically predisposed individuals, activating Th1-polarized granulomatous inflammation (1,6). Current evidence suggests that tattoos may not directly cause sarcoidosis but rather serve as environmental triggers that unmask or exacerbate a latent systemic condition. The delayed onset of reactions, sometimes years after tattooing, and the influence of additional immune triggers such as intercurrent infections, as illustrated by the case reported by Nawras et al., and vaccine exposure, as described by Albers et al., further support this hypothesis (12,21).

Treatment of tattoo-associated sarcoidosis

The management of TS is tailored to the extent, severity, and degree of systemic involvement, following a structured step-up approach similar to that used in systemic sarcoidosis (23). The treatment landscape is summarized in Table 1.

For localized, mild cutaneous disease, topical corticosteroids represent the first therapeutic step, offering targeted anti-inflammatory effects with a favorable systemic safety profile (13). Their efficacy in tattoo-associated cutaneous sarcoidosis is illustrated by our own case, in which biopsy-confirmed TS responded to topical corticosteroid therapy within one month of initiation, with concurrent improvement of systemic symptoms following adjustment of the oral treatment regimen (Figure 2). As noted above, the simultaneous increase in systemic methylprednisolone dose makes it difficult to attribute the cutaneous improvement exclusively to the topical agent. For persistent or larger lesions, intralesional corticosteroid injections (or subcutaneous injections) provide a more targeted approach, as detailed in the comprehensive treatment guide by Serup et al. (13). In selected cases with limited disease not causing significant functional impairment, a watchful waiting strategy may be appropriate, as spontaneous resolution of cutaneous sarcoidosis has been documented in certain patients (14).

Beyond topical approaches, tetracycline antibiotics—in particular minocycline—have demonstrated anti-inflammatory activity in cutaneous sarcoidosis and may represent a useful adjunctive or alternative option. Sheu

Table 1. Key references on the treatment of tattoo-associated sarcoidosis.

Ref #	Authors (Year)	Journal	Study Type	Treatment Focus	Key Findings
13	Serup J (2017)	Curr Probl Dermatol	Review	Topical/systemic/ intralesional corticosteroids; immunosuppressants	Systemic corticosteroids for sarcoidosis-related black tattoo reactions; laser contraindicated in allergic reactions
10	Serup J & Baumler W (2017)	Curr Probl Dermatol	Review/Guide	Corticosteroids; Q-switched & picosecond lasers	Sarcoidosis recognition in black tattoos critical; laser outcome unpredictable; individualized treatment required
12	Nawras A et al. (2002)	Dig Dis Sci	Case report + Review	Oral prednisone (short course)	IFN-alpha-triggered systemic sarcoidosis at tattoo site; responded to short-course prednisone
24	Sheu J et al. (2014)	Dermatol Online J	Case report + Review	Minocycline 100 mg twice daily	Rapid resolution of pruritus within 4 days; improvement of sarcoidal plaques within 1 week
15	Hirani S et al. (2020)	Am J Med	Case report	Intralesional betamethasone -> oral prednisone -> methotrexate	Step-up approach in TS; MTX added as steroid-sparing after inadequate corticosteroid response
16	Kushima H et al. (2020)	BMJ Case Rep	Case report	Systemic corticosteroid therapy	Tattoo-induced systemic sarcoidosis with pulmonary and uveal involvement; remission with corticosteroids
17	Ishimaru N et al. (2021)	Intern Med	Case report	Corticosteroids first; antiviral therapy (HCV) sequentially	Sequential approach enabled safe management of sarcoidosis and HCV
9	Kluger N (2018)	J Eur Acad Dermatol Venereol	Comparative review	Variable: observation, topical/systemic corticosteroids, immunosuppressants	Proposed TAGU (Tattoo-Associated Granulomas and Uveitis) entity; treatment guided by systemic extent and uveitis severity
18	Thrasher M et al. (2024)	BMJ Case Rep	Case report	Systemic corticosteroids; multidisciplinary management	TS with hepatic and ophthalmic involvement; specialist coordination essential
20	Dai C et al. (2019)	Am J Clin Dermatol	Literature review	Biologics: infliximab, adalimumab (3rd line); etanercept, rituximab, golimumab, ustekinumab	Infliximab and adalimumab best-supported 3rd-line options; larger trials needed
19	Paolino A et al. (2021)	Clin Exp Dermatol	Prospective cohort (6 yr)	Systemic corticosteroids -> HCQ -> MTX -> AZA/MMF	MTX remission 88% vs HCQ 40% (OR 9.8); MTX superior to AZA and MMF; steroid-sparing required in 87%
23	Melani AS et al. (2021)	Pulm Ther	Review	GCs first-line; MTX second-line; infliximab/ adalimumab third-line	>50% chance of long-term GC treatment; prednisone >10 mg/ day linked to severe side effects; step-up algorithm advocated

Table 1 (*Continued*)

Ref #	Authors (Year)	Journal	Study Type	Treatment Focus	Key Findings
11	Redl A et al. (2024)	Lancet Rheumatol	RCT (n=16)	Sirolimus (mTOR inhibitor): topical 0.1% vs systemic 2 mg/day	Systemic sirolimus improved 70% of GC-refractory patients incl. TS; effect >1 yr post-treatment; no serious AEs
21	Albers CC et al. (2023)	Acta Derm Venereol	Case report	Corticosteroids; multidisciplinary approach post-COVID-19 vaccination	Systemic sarcoidosis with tattoo involvement post-COVID-19 vaccine; novel immune trigger highlighted
30	Damsky W et al. (2022)	Nat Commun	Open-label trial (n=10)	Tofacitinib (JAK inhibitor) 5 mg twice daily x 6 months	All 10 patients improved; 6 complete responses (CSAMI); internal organ improvement also observed; IFN-gamma identified as central mediator; no serious AEs.

Abbreviations: AZA = azathioprine; GC = glucocorticoid; HCQ = hydroxychloroquine; MMF = mycophenolate mofetil; MTX = methotrexate; RCT = randomised controlled trial; TS = tattoo sarcoidosis; AEs = adverse events. All references verified April 2026.

et al. (24) reported a case of TS in a patient with pre-existing pulmonary sarcoidosis who developed raised sarcoidal plaques within tattoos present for over ten years; treatment with minocycline 100 mg twice daily led to resolution of pruritus within four days and marked improvement of plaques within one week. This rapid response, together with the favorable side-effect profile of minocycline compared with corticosteroids, suggests that tetracyclines should be considered as a therapeutic option, particularly in patients in whom corticosteroids are poorly tolerated or contraindicated.

When systemic involvement is present—or when cutaneous disease is extensive and impacts quality of life—systemic corticosteroid therapy remains the cornerstone of treatment (23). Melani et al. highlighted in a comprehensive review that more than 50% of sarcoidosis patients initiating glucocorticoid therapy will require long-term treatment, and that prolonged use of prednisone at doses exceeding 10 mg/day is frequently associated with serious adverse effects, including osteoporosis, diabetes mellitus, hypertension, and immunosuppression. This underscores the importance of early introduction of steroid-sparing strategies and regular structured follow-up (23).

Several case reports document the utility of systemic corticosteroids as the primary intervention in TS. Hirani et al.described a step-up approach in which intralesional betamethasone was followed by oral prednisone and subsequently methotrexate as a steroid-sparing agent after an inadequate corticosteroid response (15,23). Kushima et al. reported a case of tattoo-induced systemic sarcoidosis with concurrent pulmonary and uveal involvement in which systemic corticosteroids led to complete remission (16). Ishimaru et al. (17) described a therapeutic dilemma in a patient with concomitant untreated hepatitis C, resolved through a sequential approach in which granulomatous inflammation was first controlled with corticosteroids, and antiviral therapy was subsequently initiated safely. This case is of particular relevance to our patient, who also has a history of hepatitis C.

For corticosteroid-resistant or chronically relapsing disease, steroid-sparing immunosuppressive agents are the standard of care. Among available options, methotrexate (MTX) is supported by the strongest comparative evidence (19). The prospective 6-year cohort study by Paolino et al. demonstrated that MTX achieved clinical remission in 88% of patients with cutaneous sarcoidosis, compared with 40% for hydroxychloroquine (HCQ), with an odds ratio for remission of 9.8 (95% CI 2.4–40.4) (19). MTX was also statistically superior to both azathioprine (AZA) and mycophenolate mofetil (MMF) in achieving remission (OR 22 for both comparisons). Notably, 87% of patients ultimately required steroid-sparing therapy. MTX is typically administered once weekly as a single oral dose for at least two years (23). HCQ remains a well-tolerated second-line alternative, particularly in cutaneous-predominant disease. Additional agents such as azathioprine, mycophenolate mofetil, thalidomide, and pentoxifylline have also been used in selected patients with refractory cutaneous manifestations (23,25).

Biologic therapies represent an emerging option for patients with disease refractory to conventional immunosuppression. Dai et al. found that infliximab and adalimumab, both TNF-alpha inhibitors, are the best-supported third-line agents for chronic or resistant cutaneous sarcoidosis, consistent with the broader sarcoidosis treatment literature (20,23,25). Scattered reports exist for etanercept, rituximab, golimumab, and ustekinumab, though their evidence base in TS remains limited.

A particularly promising therapeutic avenue involves the inhibition of the JAK-STAT pathway. Damsky et al. conducted an open-label trial (NCT03910543) in 10 patients with cutaneous sarcoidosis treated with tofacitinib, a Janus kinase (JAK) inhibitor, for 6 months (30). All patients experienced improvement in their skin, with 6 achieving a complete response as assessed by the Cutaneous Sarcoidosis Activity and Morphology Instrument (CSAMI). Improvement in internal organ involvement was also observed. Mechanistically, tofacitinib was found to act predominantly by suppressing type 1 immunity, with CD4+ T cell-derived IFN-gamma identified as a central cytokine mediator of macrophage activation in sarcoidosis. The effect of tofacitinib suppression of IFN-gamma correlated with clinical improvement (30). These findings represent a significant advance in the pathogenesis-driven management of cutaneous and systemic sarcoidosis and suggest that JAK inhibitors warrant further evaluation in larger randomised trials, including in patients with tattoo-associated cutaneous disease.

A particularly promising novel therapeutic avenue involves the mTOR pathway. In a randomised single-centre trial, Redl et al. demonstrated that systemic sirolimus achieved clinical and histological improvement in 70% of patients with glucocorticoid-refractory cutaneous sarcoidosis, including cases with tattoo-associated disease, with a long-lasting effect extending more than one year beyond treatment discontinuation and without serious adverse events (11). These findings represent a significant advance in the management of refractory TS and warrant validation in larger, multicentre randomised trials.

Regarding adjunctive approaches, Kluger proposed the *Tattoo-Associated Granulomas and Uveitis* entity to describe cases associating granulomatous tattoo reactions and uveitis in the absence of systemic sarcoidosis (9). Thrasher et al. illustrated the complexity of TS with multiorgan involvement managed with systemic corticosteroids and close multidisciplinary coordination (18). Albers et al. (21) reported a case of systemic sarcoidosis with cutaneous tattoo involvement following COVID-19 vaccination, underscoring the importance of considering novel immune triggers.

Laser therapy has been explored for removal of granulomas or pigment deposits (14). However, evidence remains limited, concerns persist that laser procedures may exacerbate inflammation, and laser therapy is explicitly contraindicated in allergic tattoo reactions due to the risk of anaphylaxis (10). It should therefore be considered only on an individualized basis following multidisciplinary assessment.

Overall, in the absence of randomized controlled trial data specific to TS, management must follow an individualized, stepwise approach guided by disease severity, systemic organ involvement, comorbid conditions, and treatment response, with close multidisciplinary monitoring (18-28). The case presented here adds to the growing body of evidence supporting topical corticosteroids as an effective and safe first-line treatment for biopsy-confirmed cutaneous TS, although the concurrent modification of systemic therapy in our patient should be taken into account when interpreting the therapeutic response. Further clinical research—including adequately powered comparative trials—is needed to establish evidence-based treatment guidelines for this emerging and underrecognized clinical entity.

Conclusion

The relationship between tattoos and sarcoidosis remains an evolving area of clinical and translational investigation. Accumulating evidence suggests that tattoo pigments may act as environmental triggers or perpetuating factors in genetically susceptible individuals (1,6). While causality has not been definitively established, clinicians should maintain heightened vigilance when evaluating granulomatous skin lesions in tattooed patients, and a comprehensive clinical, histological, and systemic assessment is required to differentiate localized foreign-body reactions from true systemic disease (2,7). Importantly, when sarcoid lesions develop at sites of prior cutaneous trauma in tattooed patients, the contribution of scar tissue as a co-localizing risk factor should also be considered. Topical corticosteroids represent an effective first-line approach for localized cutaneous TS, while systemic corticosteroids remain the cornerstone for extensive or systemic disease. Methotrexate is the preferred steroid-sparing agent supported by the strongest comparative evidence, and biologic therapies—particularly infliximab and adalimumab, as well as the novel mTOR inhibitor sirolimus and, more recently, the JAK inhibitor tofacitinib, may represent promising options for refractory disease (11,19, 20, 29). Further research will enhance our understanding of disease pathogenesis and inform evidence-based preventive and therapeutic strategies, ultimately contributing to improved outcomes in this emerging clinical entity.

Ethic Approval: Not applicable; this is a case report.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

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